Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) MODEL EXAMINATION, 2026

THIRD SEMESTER

Skill Development Components (SDC) - MG3SDCSNP202 - ELECTROTHERAPY

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Objective Type Questions

Answer all questions, Each question carries 1 mark

1. What is current electricity? [K] / [CO1]

2. What is an electrode? [K] / [CO1]

3. Name one type of current used in electrotherapy. [K] / [CO1]

4. What is galvanic current? [K] / [CO2]

5. Define chronaxie. [K] / [CO2]

6. Expand TENS. [K] / [CO2]

7. What is Interferential Therapy (IFT)? [K] / [CO3]

8. What is phonophoresis? [K] / [CO3]

9. Name one type of ultraviolet radiation. [K] / [CO4]

10. What is contrast bath therapy? [K] / [CO4]

[1 × 10 = 10 Marks]

Part B

Short Answer Questions

Answer any 10 questions, Each question carries 2 marks

11. Differentiate between direct current and alternating current. [U] / [CO1]

12. Mention any two factors considered while selecting an electrode. [U] / [CO1]

13. What are unipolar and bipolar electrode placements? [U] / [CO1]

14. State two physiological effects of faradic current. [U] / [CO2]

15. Define rheobase and state its significance. [U] / [CO2]

16. Mention any two types of TENS and their uses. [U] / [CO2]

17. State the basic principle of Interferential Therapy. [U] / [CO3]

18. Mention two therapeutic effects of Shortwave Diathermy. [U] / [CO3]

19. What is the piezoelectric effect in ultrasound therapy? [U] / [CO3]

20. Mention two physiological effects of Infrared Radiation. [U] / [CO4]

21. State two therapeutic uses of ultraviolet radiation. [U] / [CO4]

22. Mention two precautions to be followed during cryotherapy. [U] / [CO4]

[2 × 10 = 20 Marks]

Part C

Short Essay Questions

Answer any 5 questions, Each question carries 4 marks

23. Explain the basic physical principles of electrotherapy, including static electricity, current electricity, magnetism and the electromagnetic spectrum. [U] / [CO1]

24. Describe the types of electrodes and explain the principles of unipolar and bipolar electrode placement. [An] / [CO1]

25. Explain faradic and galvanic currents with their physiological effects and therapeutic applications. [An] / [CO2]

26. Describe rheobase, chronaxie and the strength-duration curve and explain their importance in electrodiagnosis. [An] / [CO2]

27. Explain the principles, dosage parameters and therapeutic applications of Interferential Therapy and Russian current. [An] / [CO3]

28. Describe Shortwave Diathermy and ultrasound therapy with respect to principles, physiological effects and therapeutic uses. [An] / [CO3]

29. Explain the principles, therapeutic applications and safety precautions of phototherapy and superficial thermal modalities. [U] / [CO4]

[4 × 5 = 20 Marks]